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Credit - Drishti Ias

South Korea Completes Trial Voyage via Arctic Shipping Route

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Source: ET

South Korea's container ship PanStar Acro completed the country's first commercial container-ship round-trip voyage between Busan and Europe via the **Arctic** , returning to Busan after a 50-day journey.

- **About:** The trial voyage tested the **Northern Sea Route (NSR) ,** an Arctic shipping corridor running along Russia's northern coastline connecting the **Pacific and Atlantic sides of Eurasia.**
- **Voyage Details: Voyage Details:** PanStar Acro departed from **Busan, South Korea, on 22 August 2026** , carrying commercial cargo, and called at **Felixstowe (UK), Rotterdam (Netherlands) and Gdańsk (Poland)** before returning to Busan.
- **Strategic Objective:** South Korea aims to develop Busan into a **major global maritime and logistics hub** and establish regular Arctic shipping services **by 2030.**
- **Time & Distance Advantage:** The Arctic route could reduce the sailing distance between Asia and Europe by **approximately 35% compared** with the traditional route through the Suez Canal , potentially reducing transit time and improving trade connectivity.
- **Major Challenges** : Commercial viability remains uncertain due to **seasonal sea ice, unpredictable weather, specialised insurance costs, fuel consumption, limited navigation windows** and geopolitical risks along Russia's Arctic coast.
- **Environmental Concerns:** Increased Arctic shipping could raise risks of **black carbon emissions, oil spills and ecological disturbances** in the region's fragile ecosystems.
- **Strategic Significance:** The NSR could **diversify global shipping corridors, strengthen supply-chain resilience and reduce dependence on conventional maritime chokepoints.**

Read more: India and the Dynamism of Arctic Region

Facial Recognition Technology

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Source: IE

The debate around **facial recognition technology (FRT)** and surveillance evasion gained traction after suggestions emerged that creative makeup could help evade facial recognition cameras during protests.

- **Facial Recognition Technology (FRT)** : It is an advanced biometric system that identifies or authenticates individuals by mapping their **facial features from digital images or video.**
  - Rather than comparing images pixel-by-pixel, modern **FRT leverages Artificial Intelligence (AI)** to extract distinguishing patterns and convert visual traits into a numerical representation, known as a **face embedding or template.**
- **Use Cases:**
  - **1:1 Verification:** Unlocking smartphones or verifying identity at airport gates (e.g., **Digi Yatra**).
  - **1:n Identification:** Scanning crowds via CCTV to find matches in a criminal database (law enforcement use).
- **Accuracy Limitations:** FRT relies on similarity scores and matching thresholds. A positive match is not conclusive proof of identity and requires independent verification.
- **Evading FRT:**
  - **The CV Dazzle Project (2010):** Artist Adam Harvey explored using high-contrast, bold makeup and asymmetrical hairstyles to break automated pattern recognition systems.
    - This was inspired by **World War I (WW I) naval "Dazzle" camouflage** , which used geometric shapes to confuse enemy targeting rather than conceal the ships.
  - **Scientific Evasion:** A 2021 study by Ben-Gurion University demonstrated that machine-learning-generated makeup patterns (targeting highly identifiable facial regions) could bypass FRT software with up to 98% success.
- **Judicial Observation:** The Supreme Court noted that it will evaluate the deployment of FRT on the touchstone of **"proportionality,"** a key tenet established in the ***K.S. Puttaswamy* (Right to Privacy) judgment (2017) .**
  - It directed that law enforcement must strictly tie real-time FRT deployment to existing criminal data to curb broader crowd monitoring.
- **Key Concerns Surrounding FRT in India:** While the **Digital Personal Data Protection (DPDP) Act, 2023** regulates personal data, India lacks a dedicated statutory framework governing police FRT deployment, particularly concerning watchlists, matching thresholds, data retention and independent audits.
  - Algorithms are prone to "false positives," particularly concerning race and gender, which can lead to wrongful arrests and bias against marginalized communities.
  - The fear of being identified, profiled, and added to state databases can deter citizens from exercising their **fundamental right to peaceful assembly (Article 19).**

Read more: **Regulating Facial Recognition Technology in India**